

Gamification and Serious Games for Cybersecurity Awareness and First Responders Training: An overview

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Abstract: Gamification has been characterized as the usage of video game elements in non-game environments. Gamification in cybersecurity awareness training can lead to more engaging interactions, greater pleasure, and increased security. Serious games are becoming increasingly popular since they allow learning in a natural setting. Although many of the principles covered in cybersecurity awareness training are universal, many forms of training fail because they rely on rote learning and do not require users to think about security concepts. The primary goal of this research is to conduct a review of the literature on cybersecurity awareness training methodologies and frameworks, as well as serious games that may be used to raise awareness among people, organizations, and first responders.

Keywords: cybersecurity; awareness; serious games; first responders; disaster; training.

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1. Introduction

Cybersecurity concerns have grown significantly enough in the contemporary digital era to garner widespread attention, with significant cyber-attack incidents making news across different media outlets. Critical infrastructure from all sides of the sector has been a prominent target of cyber attacks in recent years. According to technical studies, attackers used several tactics to obtain access to various computer systems, including attaching malware to third-party programs, emails and websites. As a result, the terrorists were able to carry out sabotage activities that may have affected the electricity supply in numerous European countries and the United States. The success of such attacks was frequently dictated by user ignorance and a lack of rigorous staff training[1]. According to an IBM report from 2022, human errors were responsible for 25% of security breaches in industrial firms that year. [2]. In 2021, the most common primary cause of data breaches was phishing and related attacks, as well as business email compromise. The malware was the third most common cause of data breaches, trailing only ransomware. According to ITRC, ransomware attacks will overtake phishing as the leading root cause of data compromises in 2022 if current trends continue [3]. All of these studies and reports show that user awareness and training are essential to the success of many cyber-attacks. There has been a wealth of research conducted in recent years on the creation and implementation of frameworks

for cybersecurity workforce education and awareness. Despite this, a consensus on the most effective overall solution has yet to be reached. Many studies have highlighted the advantages and disadvantages of various training options, but there is still debate on which strategies for cybersecurity awareness training are the most effective. This lack of agreement and the urgent need for a well-prepared CS workforce prompted more research into current solutions for CS training for Critical Infrastructure protection. In recent years, the behavioural element of cybersecurity has received increased attention. People can increase the security of their devices and organizations through their activities, but they can also frustrate the successful implementation of security in these areas [1].

The advancements in internet, mobile technologies and in general communications have opened up new chances for individuals and organizations in both the public and private sectors to conduct transactions online. This has led to the generation of large amounts of data, which is often stored on servers worldwide. Those responsible for handling sensitive information, in example banking data, have allocated significant resources to enhance cybersecurity measures by bringing on board experts, implementing robust security policies, adopting new technologies, and providing training for security professionals. However, little attention has been given to raising awareness among customers or users about cybersecurity, making them vulnerable to cyber attacks. As a result, cybercriminals are focusing on exploiting the human factor, using tactics such as identity fraud, ransom attacks, phishing and social engineering to target victims through online chat forums, emails, and other means. With the increasing number and complexity of cyber attack methods, types, and technologies targeting vulnerabilities in users, the role of human factors in cybersecurity awareness and management becomes more critical. To combat these cyber attacks that exploit human weaknesses and protect information assets, it is essential to develop cybersecurity awareness programs that educate users on their roles and responsibilities. Research has shown that a lack of knowledge and awareness about cybersecurity threats is a significant contributor to the rise in attacks. This is particularly true in academic institutions where many users are students, who may inadvertently contribute to data breaches and digital misconduct due to their lack of understanding about cybersecurity and the consequences of cybercrime. University students are particularly susceptible to cyberattacks because of their extensive usage of various online applications and social media platforms, which increases the risks associated with their daily activities. Unfortunately, many students lack knowledge about the basics of cybersecurity and do not have the necessary knowledge to protect their devices from malware, viruses, and fraud. As a result, improving cybersecurity awareness in academic institutions in terms of understanding network security and protection methods has become critical. There is an urgent need to apply a standardized strategy to raise cybersecurity awareness among college students, increasing their understanding of these concerns and educating them on how to protect themselves from future intrusions [4].

Cybersecurity is a multifaceted, multidisciplinary global phenomenon. A cybersecurity breach can have low to moderate consequences, such as data theft and manipulation, or it might involve seizing control of systems and causing bodily harm. [5] Serious games are getting increasingly popular since they allow learning in a natural setting. Although many of the principles covered in cybersecurity awareness training are universal, many types of training fail because they rely on rote learning and do not require users to think about security concepts. In studies spanning educational environments, the learning impacts of serious games have been inconclusive. One of the suggestions is to ensure that gaming and learning objectives are in sync. Because the learning purpose and consequences are unknown, determining whether or not a game aids end-user learning is exceptionally challenging. [6]. Serious game applications in cybersecurity extend from training exercises to safety and security games, which are a viable alternative to traditional safety training and allow learners to explore diverse scenarios before experiencing them in their daily lives. [7].

Security Education Training and Awareness can help companies endorse the thoughtful-

ness and accessibility of their resources. Securing systems from cyber-attacks is one of the most challenging concerns that has arisen as a result of technological improvements. There is a specific need to adopt comprehensive infrastructure security, starting with the grassroots and local government. With the development of hacktivist groups such as Anonymous, whose sole purpose is to cripple the information systems of various governments, cybersecurity has become an essential factor in the teaching of information systems. Each organization's information system analysts are responsible for the cyber education of its employees, ensuring they are aware of the risks that exist in cyberspace and making informed decisions that are in the best interests of the organization's security.

2. Methods

A thorough literature review was used in this investigation. A systematic review aims to locate relevant studies, appraise their contents, uncover common themes, and critically summarize their findings. To begin, a research plan was created that linked the research purpose, keywords, and a set of inclusion and exclusion criteria. The research aims were to conduct a literature review on gamification to increase emergency preparation processes' efficiency through community participation. This study aimed to assess the applicability and effects of gamified features in generating community engagement in emergency preparation activities. Based on the objectives listed above, it was agreed that the core keywords would be 'gamification', 'serious games', 'emergency', 'cybersecurity', 'disaster', and 'first responders'. The inclusion requirements were peer-reviewed research articles written in English. The review was conducted with the help of an academic search engine that connects databases such as Science Direct, Scopus, Web of Science, and the Directory of Open Access Journals. From June 2022 to September 2022, the search was undertaken for journal papers published between January 2010 and September 2022. The initial thematic search was conducted using the Boolean search line ((Title/Abstract contains the term/s "gamification" OR "serious game") AND (Title/Abstract contains the term/s "first responders" OR "disaster")) to identify studies focusing on gamified applications for disaster emergency. The term gamif* was truncated to extend the search to synonyms such as 'gamification' and 'gamified'. After reading the full texts, the papers were reduced to 66 to confirm that they fit within the research topic. The approach concluded with a critical evaluation and documentation of the findings in the form of a literature review paper. Additional publications that did not match the selection above criteria were included in this procedure to better grasp the study background and conclusions.

3. Game-based approaches

Gamification is used to increase students' interest and involvement in their studies. Through a gamification-based training program, game components are used to give real-life work settings. Gamification-based training programs are designed to achieve organizational goals by aligning with strategic objectives. According to research, gamified training programs provide a unique environment of learning that piques learners' attention and boosts their engagement with the information. Developers include a variety of training components in their games, such as leader boards, badges, challenges, group competition, levels, storytelling, points, feedback, avatars, social points, feedback, incentives, likes, and achievements, to encourage learners to take charge of their learning journey. These features provide information about the training materials and increase participation in the training program. However, from a research standpoint, studies on the effectiveness of gaming components and learning outcomes remain scarce. According to studies, the most popular gaming features used in gamification-based training programs include points, leader boards, challenges, ranking, and scores. Learning Gamification and theories Education, psychology, sociology, and neuroscience have all contributed to the development of learning theories. In the context of gamification and its impact on student learning outcomes, research has extensively used self-direction theory, flow theory, and goal setting

theory. Gamification has also been examined using cognitive evaluation theory, behavior reinforcement theory, social comparison theory, rational choice theory, self efficacy theory, constructivist learning theory, and the technology-enhanced training effectiveness model. The most widely utilized theories are the self-determination theory, the flow theory, and the goal-setting theory [8].

- The self-determination theory focuses on characteristics that contribute to an individual's sense of autonomy, competence, and ability to adapt to training programs. The self-determination theory clearly specifies the aspects that should be incorporated in any gamification program to drive learners to interact with the course and take charge of their learning journeys.
- According to the flow theory, learners are driven by demanding learning situations, which promotes a flow toward more learning. The flow theory of learning highlights the components that might keep the learners' learning flow running.
- The goal-setting theory assists in identifying the characteristics that inspire learners to attain learning objectives. The goal setting theory advocates setting learning goals and then constructing gamification content to drive learners to achieve them.

Security Awareness and Training (SAT) programs are frequently implemented to mitigate the risk associated with end-user insecure behavior. However, studies have called into doubt the effectiveness of SAT programs in terms of improving end-user behavior. In this context, we investigated the possibility of using the concept of gamification to improve motivation and learning results. Based on discussions with security specialists and a workshop with normal employees from two firms, an interactive SAT prototype application was created. Employees tested the prototype in a second workshop. The goal of this research was to see if, and maybe how, the use of gamification - that is, using game concepts and techniques to engage and inspire individuals - could improve the learning environment for SAT programs. [9] gave an evaluation of a prototype that uses gamification to teach security awareness. They conducted an empirical study using data gathered from interviews and group workshops. The goal has been to determine how employees might be best encouraged to participate in security training using gamification. [The study discovered that many of the issues the SAT is currently facing are issues that gamification is supposed to tackle. The end user is important to the gamification design process. Through the goals that were agreed upon as shared between company and employees, the study revealed four reasons why employees were interested in SAT. It is essential that SAT programs be designed to achieve those objectives. The more self-determined the training, the more motivating it will be, it was discovered. Gamification has the potential to provide consumers with personalized material based on their talents and role in the firm. Current training courses, according to popular perception, are excessively long. An important consideration was the use of small, concise exercises rather than hour-long e-learning courses. It was discovered that this feature can provide two important outcomes: (1) give users a sense of autonomy over the training, and (2) improved learning outcomes due to the spacing effect, and that the threshold for starting the training is lower when it does not require planning—it can be done on the go, when there is time, provided that the learning platform supports this. Finally, the goal of an SAT program is to instill good security behavior in employees. Our findings suggest that gamification has the potential to be used in SAT programs, particularly in areas where current SAT efforts are thought to fail. However, there are significant pitfalls to avoid when designing such applications, and more research on the long-term effects of a gamified SAT application is required [9]. The interaction between users and various technologies has grown rapidly. The use of various applications for various purposes has been introduced into people's daily habits. The provision of appealing services is an important aspect that should be taken into account during their design. Gamification facilitates this, while game elements create a more entertaining and appealing environment. At the same time, because these technologies collect and store users' information, security and privacy must be considered in order for these technologies

to provide a minimum level of security and protection of users' information. Users, on the other hand, should be aware of their security and privacy so that they can recognize how to protect themselves when using gamified services. The relationship between privacy and gamified applications is discussed in this work, with implications for both software developers and users, highlighting the importance of not only designing privacy-friendly systems but also educating users on privacy issues through gamification.

Game-based learning is based on the principle of teaching through repetition, trial and error, and accomplishment. This approach is similar to that of video games, in which players start at a slow pace, gain experience, and eventually become proficient in managing more challenging levels. Instructive games have become a recent resource that is utilized to support and promote students' learning experiences. Game-based learning has been shown to lead to an increase in enjoyment and engagement in higher education, especially when connected to the curriculum and learning objectives, which can significantly enhance deeper understanding and retention. Additionally, game-based learning has been applied with success in various areas, such as healthcare, military, and management, and has been demonstrated as an effective training method. Game-based methods for cybersecurity education have also shown to improve the effectiveness of cyber security learning and training.[4]

3.1. Raise cyber security awareness

Gamification has been defined as "the use of game-play elements for non-game purposes," and it is applied in many fields including education, business, and medicine. One of the most essential aspects of learning through this approach is that participants actively and seek solutions to problems by confronting them. Problem resolution through gamification has also been identified as an essential benefit of employing games in education. A problem-solving mechanism based on a game-based strategy allows for information acquisition as well as application throughout the learning process [10]. Kaspersky Industrial Protection Simulation (KIPS) is a cyber security exercise by team battle that is aimed to improve analytical skills about cyber security concerns and the risk of the most recent computer system in operation. Executive managers, including company managers, department heads, and information security administrators, are educational targets. The goal of KIPS is to maximize profit while preserving confidence after exposure to a succession of unanticipated cyber threats. The goal is to build and implement cyber defense tactics by picking the best-suited strategy from a preliminary list of cyber security countermeasures.[11]

Because technology is always evolving, there is a continuing need to renew police officers' cyber security training to guarantee that the UK remains a safe place to live and do business. Rather than traditional classroom-based training, our project evaluates the effectiveness of cyber security delivery through the use of games-based learning to simulate cybercrime and give incident response training. Computer games provide aesthetically beautiful, interactive, and immersive environments that enable for the exploration of complicated problem areas for both amusement and serious reasons. Existing research shows that games can be entertaining for a wide range of users and are an effective technique for boosting activity participation. It has also been demonstrated that games allow users to interact with an environment that simulates the real world and have a good impact on motivation, enjoyment, pleasant moods, and happiness, which can increase learning and knowledge retention. [12]

CodeHunt is a platform that gamifies cybersecurity training. Platform for interactive educational games. To familiarize participants with the concepts and techniques, first introduction texts are employed. An gradual method is chosen, in which new subjects and tasks are introduced one by one. It is advised that additional support materials be included to further support participants. Adaptability, analysis, architecture, evaluation and feedback, design, game mechanics, learning outcome, and pedagogical concerns are among the important ideas covered.

Cybersecurity is a multifaceted worldwide phenomena that presents governments and the commercial sector with complicated socio-technical concerns. As technology evolves, the types and amounts of cyberattacks effect various users in different ways. The bulk of known cyberattacks are the result of human error. Despite the fact that it is both knowledge- and environment-dependent, research suggest that improving users' cybersecurity awareness is one of the most effective preventative measures. However, because of their intangible character, socio-technical interdependence, continual technological evolutions, and unclear impact, comprehensive techniques for better communicating and combating cyberattacks are difficult to propose. Industry has primarily focused on creating a culture of risk awareness within organizations through research, while academic research tends to focus on assessing students' attitudes and behaviors after incorporating cybersecurity awareness into specific courses, rather than making it a fundamental part of an academic institution's curriculum. This is despite the importance of equipping all graduates with the knowledge and skills to combat cyber threats as a core mission of academic institutions.

3.1.1. Frameworks, methods and platforms

[4] provides a conceptual Cybersecurity Awareness Framework to guide the development of tools to increase graduates' cybersecurity awareness in any academic institution. This framework aims to enhance the integration, development, delivery, and evaluation of cybersecurity knowledge into a university curriculum, across different disciplines and fields of study. This will result in a higher level of awareness among all university graduates, who will be part of the future workforce. This framework can serve as a guide for academic institutions in creating or revising policies and procedures for the development and evaluation of cybersecurity awareness.

A unique paradigm for cybersecurity training was presented with three primary objectives: I training should be launched by emerging risks and offered first to the most vulnerable individuals (ii) the process must be dynamic (iii) training results must provide actionable intelligence. It links cybersecurity awareness training and cyber threat intelligence through ontology and relationships. The proposed solution is feasible and scalable using IBM Watson Knowledge Studio. The project produced a manually annotated corpus of 100 threat reports and 127 cybersecurity training evaluations, dictionaries for pre-annotation, and an annotation model with 66% entity marking coverage and 75% accuracy [13].

Before creating customised awareness content, you must assess the audience and issues that will appeal with them. Some content may be suitable for the general population, but age, education, and technical skills must also be considered. Understanding your audience's demographics is crucial to effective user awareness. For that group and topic, the message should be approachable, realistic, and memorable. Simple, direct, and engaging. Once you know your audience, issues, and goals, you may gather resources and content ideas. Keeping a repository of trustworthy sources for material production can reduce the effort, time, and resources needed to create awareness and training materials. Cybersecurity awareness spans borders, sectors, and organizations; there are many publicly available materials that may be adapted to different organizations and audiences. After collecting materials and brainstorming, you can organize and personalize the information [14].

The objective of this research is to develop a cybersecurity awareness training tool that allows users to grasp security concepts through gameplay. To achieve this, a game was designed using the Activity Theory-based Model of Serious Games (ATMSG) and conducted a study to examine its effectiveness. The results suggest that incorporating a story-based gameplay can aid players in enhancing their understanding of cyberattacks and the measures to counter them. [6].

Cyberattacks on industrial control systems have increased in recent years. Proper software development helps maintain these systems secure. Developers must follow secure coding principles and best practices to create secure software. This study gives a framework for software developers to raise security awareness. The authors also present a virtual coach,

an interactive game component that uses a rudimentary AI engine based on interview laddering. A study of real-world participants suggests the built platform is well-received. Preliminary results reveal that participants enjoy the platform and it raises awareness about secure coding best practices. After internal software clearing, the built platform will be released. In future work, the authors want to examine additional aspects that help software developers comprehend attacking susceptible code. [15]

To raise cybersecurity awareness, a chatbot with a knowledge base on cybersecurity was proposed. Users can use the chatbot to learn the definitions and concepts of various cybersecurity terms. Users can use the chatbot's self-quizzes to test their knowledge of various cybersecurity topics. It also advises users on cybersecurity issues such as identifying and dealing with phishing emails. The majority of users in a survey of twenty agreed that our chatbot is simple to use and can raise their awareness of cybersecurity issues. [16] CSC blends serious gaming concepts with cybersecurity and secure coding guidelines to enhance software developers' secure coding awareness. Industrial environments have leveraged these cybersecurity events successfully. The authors' cybersecurity awareness tool, Sifu, automatically assesses issues for compliance with secure coding rules and uses AI to deliver solution-guiding tips. Sifu's features allow for remote (online) learning during social distance. Four online real-life CSC events examined Sifu-based CyberSecurity Challenges. Three studies suggest that Sifu's CSC events are effective at educating software engineers about secure coding. The results demonstrate that participants enjoy utilizing the platform to promote awareness about secure coding best practices. Sifu's challenges have good ratings, showing that software engineers agree on their quality. Sifu's feedback scores are high. Work-related considerations, developer happiness, and user experience also received high evaluations.

The major goal of an awareness program is to inform users about cyber hazards in order to impact user behavior, as well as to aid the user in making the appropriate choices when navigating the Internet or interacting with computers as part of their everyday responsibilities. Users must be aware of common fraud and phishing methods, as well as the fundamental techniques used by malevolent actors to deceive naïve users into executing an unexpected behavior. A well-designed cybersecurity awareness program should promote organizational learning while also supporting the entire company objectives from a security standpoint. Users will lack the background or abilities to recognize prevalent hazards to the information systems on which they operate if they do not participate in organizational learning [14].

A concept map from LOD was proposed that can be used for cybersecurity awareness training. This is accomplished by querying useful data from the LOD database DBpedia with SPARQL to obtain related data nodes. The Page Rank Algorithm is then used to determine every node's significance. The more critical data nodes are prioritized for cybersecurity awareness training. This work contributed to the development of training concepts through a dynamic formation of a concept map from the LOD database DBpedia, which can be updated in real time. [17]

Kahoot is an educational tool that utilizes a game-based approach, enabling educators to design and distribute interactive quizzes and games with ease. It makes assessment and evaluation more streamlined through the use of reports, while also making the learning experience more engaging and enjoyable. The Kahoot platform offers a vast library of over forty million pre-made games on an array of subjects for users to choose from. [4]

3.1.2. Organizations's side

With the tremendous advancements in technology, severe cybersecurity issues have arisen. In people's daily lives, there are numerous and diverse sorts of cyberattacks. Giving people cybersecurity awareness training is quite important. However, because there are so many cybersecurity materials in the micro-world, determining what themes to train people's awareness in cybersecurity comes first. This is becoming a big worry for orga-

nizations as the frequency of security breaches caused by workers' noncompliance with security regulations rises.

Many studies on cyber awareness have concluded that an effective awareness program enhances staff security behaviors. The general consensus within the information security community is that people are the weakest link in a security chain. One reason for this is that people frequently act in ways that contradict what they intend to do. Another reason is that they frequently fail to adhere to security policies. However, no detailed findings appear to inform practitioners on how to encourage positive security behaviors and what considerations to take into account when creating an awareness campaign. Security awareness practitioners are security professionals that, as part of their job, plan and deliver cyber awareness programs to promote, educate, and drive the adoption of security best practices and technologies. The beginning point [18] for fighting cyber-attacks is the level of knowledge employees have about the cyber dangers that face their organization, their systems, and themselves. Employees are provided with informal training to increase their understanding of risk and security. The majority of breaches are the consequence of employees failing to follow the organization's security standards. [5]

Many employees have been recognized as having a rudimentary awareness of their organization's information security policies, and even if they do, there is no guarantee that they will comply with them. Awareness is the first step toward behavioral modification. Organizations must invest in modifying employees' behaviors so that they are consistent with their information security standards in both their professional and personal activities. Whoever is in charge of the awareness campaign must ensure that the operating environment of their audience is taken into account, including technology, roles, limits, and goals. Cybersecurity awareness programs should not just focus on the organization but also on the personal surroundings. It is critical that they provide reasons why individuals should follow security rules and arouse sentiments of well-being, rationality, and logic. Understanding what motivates people to engage in good security behaviors is critical because it allows practitioners to assess if their present approach needs to be modified. [5] A study provides a Competence Developing Game (CDG) for cybersecurity awareness training in enterprises. GHOST is an innovative technique to doing cybersecurity awareness training for end users in businesses. To match business and employee expectations, such training should meet at least seven standards. This work demonstrates how utilizing the CDG concept gives the resources required to construct a suitable game design. To save time, a GHOST training can be conducted at the workplace. The organizational overhead is decreased because substantial preparation is not required. Both elements contribute to lower training expenses. Employees with no game experience can engage in the training because of its sophisticated empirical interaction mechanisms. Furthermore, this interaction method contributes to GHOST's appearance and feel of an entertainment game. The turn-based, business game-inspired game design enables a continuous training cycle, which is not possible with computer-based training. Furthermore, the societal significance of - and rising attacks on - IT systems leave no doubt about the underlying problem's real-world implications. Following the evaluation of the GHOST aftereffect, GHOST will be employed in real-world cybersecurity awareness trainings. [19]

3.1.3. Raise cyber security awareness using Serious games

Cyber attacks are becoming more sophisticated and numerous, requiring organizations to constantly adjust management techniques for cyber security risks. Organizations are investing in professional training courses for their staff as a critical risk mitigation policy to enhance understanding about cyber threats and related defenses. Serious games have arisen as a novel strategy that, by offering a joyful setting in which players learn and apply cyber security topics through the game, can supplement instruction-led or computer-based security training. They suggest Riskio, a tabletop game designed to raise cyber security awareness among employees with no technological expertise. Riskio is an active learning

environment in which participants gain knowledge of cyber security assaults and defenses by acting as both the attacker and defender of important assets in a fake organization. We ran a series of trials with employees and graduate students to determine Riskio's perceived usefulness in raising cyber security awareness. The review demonstrates the game's usefulness and highlights trade-offs in terms of design and playing experience that can allow the game to better meet different audiences. The game's evaluation revealed that employees are more confident than students that Riskio can raise their understanding of cyber security topics. Employees appreciated the game principles and dynamics, and they could apply the game board scenario to their own organization, which could explain the difference. The game board is an important component for representing the intricacy of realistic scenarios, yet it may need to contain various information depending on the audience. [20]

3.2. *Serious Games for First Responders training*

Games have been developed to enhance the way technical information is conveyed to non-technical audiences, with the goal of raising flood awareness and empowering communities to prepare for flooding through promoting cooperation with local government and building citizen capacity in critical infrastructure. These games serve as educational tools to engage and entertain participants while providing a learning opportunity and introducing relevant topics. Specific skills such as spatial thinking and crisis response are emphasized in the gameplay. Some games also focus on collecting data on players' willingness to pay for probabilistic flood forecasts and their ability to understand decision-making in uncertain conditions. The games were also designed to validate existing theories and models developed by researchers, as well as to test hypotheses through data collected from players' gameplay actions. Evaluation is necessary to determine the effectiveness of the game and to support its ongoing development. However, there is a tension between player success and game success, with objectives and goals often focusing on one over the other. How can gamification be used in disaster response planning? (c) Gamified applications help shape a new culture—i.e. gamified culture—that supports smart disaster emergency planning practice by creating new opportunities for community engagement and raising disaster awareness among the community in virtual environments.

3.2.1. *Cybersecurity training*

[21] proposes a framework for creating CS serious games. The framework is step-by-step: analysis, design, development, game evaluation, deployment, player evaluation. The writers collected relevant CS abilities from internet frameworks to include in serious games. [22] proposes CyberCIEGE: a security-training video game. CyberCIEGE is a simulation engine, scenario definition language, scenario production tool, and video-enhanced encyclopedia. Each security scenario is unique. Some IT scenarios are job-specific. [23] proposes "Indra," an advanced CS simulator. Indra improves prevention, detection/reaction, forensic analysis, and attack. forensic analysis, cyber-defense, cyber-attack, and cyber-warfare are supported. Serious Game-based Approach to CS Concept Learning Raman et al. Two control groups were tested, one using game-based training and the other without. First group had better CS knowledge and effectiveness. Cyber security education is a new profession with numerous obstacles, and demand is rising. Serious games and game-based learning are promise for cyber security training. Cyber security game-based techniques lack standardized methodologies and design standards due to the newness of game-based learning. In [24], the authors present COFELET, a framework for cyber security serious games. COFELET promotes game-based techniques to improve cyber security education pedagogy by incorporating modern learning theories and creative teaching approaches. COFELET envisions a sustainable lifecycle of learning, updating, and reinforcing knowledge and skills through multidisciplinary approaches based on strong cyber security techniques, models, and tactics. The COFELET framework promotes game-based learning, serious

games, cyber world simulations, and hacking activities.

The growing number of cyberattacks is being fueled by a lack of awareness among users about unsafe online practices, such as visiting insecure websites, ignoring warning messages, and contacting with unauthenticated sources. Game-based learning can help improve cognitive development and conceptual understanding, but there are few mobile games that teach users about cyberattacks and how to prevent them. This work details the creation of an augmented reality (AR) game, called CybAR, aimed at increasing cybersecurity awareness and knowledge. The game is a mobile app that uses AR to teach cybersecurity concepts and demonstrate the real-world consequences of actual attacks. The design and evaluation of the application are also discussed. CybAR helps players understand cyberattacks and vulnerabilities, according to the results.

PeriHack is an educational tool to gain overall cybersecurity awareness from both a red and blue tactical point of view. The former can gain a better understanding of how a cyber attack can start and evolve, while the latter can appreciate the struggle of managing a multitude of possible weaknesses with limited resources, thus the need to make decisions based on perceived priorities and limited information. This work discusses how "PeriHack" was designed [25]. PeriHack is a board and card game for two players or teams that mimics a red-versus-blue battle (defenders). The game requires players to investigate a sample network for vulnerabilities and chain assaults to exploit technical and social engineering exploits. In addition, it replicates budget limits for the blue team by providing restricted resources to examine and prioritize significant risks.

In [26] a combination of two educational artifacts is proposed in order to solve the lack of efficiency in information security awareness training. The first one is an interactive for learning and testing the acquired knowledge on cyber security and hygiene practices. Quizzes, crossword puzzles, multimedia such as video, and "complete the word" games are utilized to enhance the learning process. The second one is an interactive board game designed to measure the understanding of the subject. Players are challenged to move forward in the board by correctly answering questions about different personality styles of hackers.

Due to their convenience, rapid feedback, appeal, and ease of simulating innovative and numerous assault situations, serious games are widely used in education, information systems, and information security. This research presents a serious games model to improve computer users' cyber security awareness. This study focuses on strategically deploying serious games to educate, train, and inform organizational users to identify, mitigate, eliminate/mitigate, and report social engineering threats launched by APT vectors that target organizational user weaknesses. Instead of typical security awareness training, they offer a scalable, interactive, and engaging performance-based gaming environment [27]. SherLOCKED was designed after reviewing serious games and gamification ideas. SherLOCKED, a 2D top-down detective adventure serious game, was proposed in this post [28]. We built a cybersecurity serious game for training because information-only campaigns are ineffective. Through the notion of planned behavior, the effectiveness of this game was compared to a non-cybersecurity game with or without cybersecurity knowledge. The cybersecurity game led to higher self-reported attitudes, behavioral control, intentions, and action than the non-cybersecurity games. For subjective norms, we only identified an effect without extra information[29].

3.2.2. First responders' training

As the number of global disasters rises, many mitigation strategies have been proposed. Appropriate drill scripts help first responders in a genuine emergency. Creating a drill script is time-consuming, error-prone, and expensive. First responders need time-dependent, region-specific drill scripts to see different disaster situations for improvement. An ontology model incorporating Semantic Web Rule Language (SWRL) features is presented to help Taiwan's drill script creation. Drill script designers must prepare a Taiwan Earthquake Loss Estimation System input data collection. Then, a drill script following

Table 1. Cyber Security training approaches

Reference	Domain	Description
[21]	Cyber security training	A serious game for CS training
[22]	Cyber security training	An interactive video game for security training
[23]	Cyber security training	An advanced Simulator for CS training
[24]	Cyber security training	A Conceptual Framework for e-learning and training
[25]	Cyber security training	PeriHack is a board and card game simulating the struggle between a team of attackers and a team of defenders
[26]	Cyber security/hygiene training	An interactive book and board game for optimizing learning procedure and understanding in an entertaining way
[27]	Cyber security training	A serious games model for organizations to substantially enhance computer users' cyber security awareness
[28]	Cyber security training	SHERLOCKED is a 2D top-down detective adventure serious game to support further engagement
[29]	Cyber security training	A cybersecurity serious game applicable for CS training

pre-defined principles can be integrated with Unity, a serious gaming platform, to depict earthquake-related events [30]. Blind ingress into tight locations often kills or injures rescuers. Emergency drills protect rescuers and improve rescue outcomes. VR serious games are used for emergency drills [31]. This work created a VR-based restricted space rescue SG (CSRVRSG). The rescue process in a confined-space situation is recreated in 3D to create a low-cost, efficient VR SG experience for emergency drills. Confined space and regular personnel can test CSRVRSG. This game is well-designed and relevant, and VR Serious games can increase participants' confined space rescue knowledge[31].

Lack of restricted space-specific skills and expertise leads to significant rescuer casualties. Desktop, real, or simulated drills can help. Risky and expensive drills are rarely utilized. Frequent desktop drills may be unrealistic because they focus on organizational coordination rather than rescue abilities. Simulation drills are cheap and low-risk, yet unrealistic. R SG can improve the realism, applicability, and training efficiency of simulation drills. After training, participants' knowledge of confined space rescue improved dramatically, and CSRVRSG is especially effective for newcomers. VR SG is an innovative tool for emergency response training [31]. Feng et al. [32] used a VR SG to improve seismic behavior and evacuation readiness. VR increased study participants' knowledge of BR and PEE. G. Burigat [33] used a VR SG to instruct participants on aircraft spatial awareness for effective evacuations; VR SG-trained participants evacuated the craft faster than safety card-trained participants.

Simulation-based training improves skills and knowledge. Virtual reality technology has been used in simulation largely for training, to enable learning while conducting risky or impossible-to-simulate activities. They demonstrate a simulation-based earthquake firefighter training developed with Italian firemen. The suggested training technique uses virtual reality and a novel interaction and gaming concept to train first-responders. This research aims to train first responders in a safety-critical scenario and compare it to document-based training. A model is needed to create an appropriate training by giving risk-management information. This can boost control room research. The student interacts with items and does tasks on a head-mounted display. Virtual reality improves training effectiveness, according to tests. Trainees have a greater perception of the scene, which leads to a speedier response. The proposed training approach can assist firefighters manage dangers. Despite cutting-edge VR technologies for improving UX, various problems limit broad deployment. VR usefulness is still being studied[34]. The work in [35] introduces a Serious Game that will be used to teach Catalonia's fire fighters. The game is played in a virtual setting free of genuine hazards, saving on materials and supplies. The system exposes the student to emergency circumstances that he/she must address using professional firefighters' tools and following each case's protocol. You can interact with virtual agents

to assign tasks and get emergency information. All user actions are tracked and scored at the end of each game based on sequence and time. Feedback is given to improve a player's future decision-making.

Hotzone [36] is a video game-based simulation to prepare first responders for chemical and hazardous materials crises. In his game, the instructor can control the wind, temperature, precipitation, type and position of dangerous materials, and location of victims. VR/AR/MR, sensor, cave, and somatosensory technologies are employed to create the gaming world as realistic as possible. Xu et al. [37] created a VR-based game to help crane operators learn how to handle railway accidents. The FLAIM Trainer [38] is a firefighting simulator that delivers an immersive virtual environment for safer firefighter training. First to integrate VR and haptics.

Single-player serious games are common. When teams train together in a virtual environment, the immersive aspect of serious gaming and VR may be strengthened. This article discusses the technical hurdles of constructing a multiplayer counter terrorism training serious game using virtual reality, touch screen interfaces, and a realistic crowd simulation. Each technical developer has a set of tools that are continuously incorporated, piloted, and upgraded during the development cycle. Piloting with first responders allows iterative enhancements that satisfy end-user training needs. Building a multiplayer training game that simulates real circumstances realistically and allows virtual reality interaction has several technical obstacles. Through efficient and effective planning during development, the AUGMED platform built on the strengths of using different input methods to ensure a complete training experience regardless of the learning device.

[39] describes a virtual reality framework for teaching first responders and digital forensic investigators. Training and evaluation are the framework's two elements. The framework allows training and evaluation of crime scenes to be easily customized by assigning virtual objects various functions and behaviors. The training module provides methods, pipelines, and task execution sequences. After training, the person can interact with a virtual reality crime scene by collecting evidence. This method evaluates the trainee's performance and the training process. The evaluation framework captures insufficient evidence, task execution sequence, and job completion time. The trainee or supervisor can evaluate the collected data. A basic, adjustable scoring technique was added to the framework to give the student timely performance feedback. VR training has gained popularity in numerous fields, including firefighter education. Immersive VR technology, especially for skills training, is hesitantly accepted. This study examines the first-time experiences of 19 firefighter-students, 8 instructors, and 7 veteran firemen using an immersive VR technology to simulate fire extinguishing. The technology replicated fire, smoke, suit heat, and hose pressure.

3.2.3. Disaster Management

As the importance of experiential learning in disaster education has expanded, serious games have gained prominence. Game-based learning focuses on emotions, peer interactions, communication, and active cooperation. Computer or video games are frequently used in student-centered education programs as computational, visual, and virtual instruction improve. [40] develops a serious game-based learning package for catastrophe avoidance. Education is an effective disaster control tool. Traditional teaching approaches aren't enough to pass on expertise and spark student enthusiasm. Serious game-based learning is a promising tool for catastrophe prevention education. This study combined the serious game Battle of Flooding Protection with Kolb's Experiential Learning Cycle to boost students' interest in learning, self-awareness, and desire to participate in disaster-related citizen actions. The Battle of Flooding Protection learning package improved students' disaster prevention skills, learning interest, self-awareness, and civic duty.

Serious games can increase evacuation capacity and conduct. Serious games have an instructional objective and aren't played for fun. Medical training, mental healthcare, emergency decision-making, earthquake and fire evacuations all use serious games. Fire

Table 2. First Responder training approaches

Reference	Domain	Description
[30]	First responder training - Earthquake drills	A serious game platform displaying all earthquake-related events in a virtual environment
[31]	First responder training	A confined space rescue serious game based on VR technology
[32]	First responder training	A VR SG to enhance earthquake behavioral responses and post-earthquake evacuation preparedness
[33]	First responder training	A VR SG to train participants on spatial knowledge of an aircraft for effective evacuations
[34]	First responder training - Earthquake drills	A simulation-based firefighter training for an earthquake scenario
[35]	First responder training	A serious games to train fire fighting students at the Public Security Institute of Catalonia
[36]	First responder training	An instructor-based simulation game to train first responders on chemical and hazardous materials emergencies
[37]	First responder training	A VR-based game to learn about how to deal with railway accidents
[38]	First responder training	FLAIM Trainer is a firefighting simulator for training in realistic fire emergency situations
[39]	First responder training	A virtual reality framework for training first responders and digital forensic investigators

evacuation is crucial. [41] proposes a web-based serious game for virtual fire evacuation drills to increase occupant safety. A stand-alone virtual drill system was designed as a prototype for the serious game. On this premise, the serious game's system framework, including the database, front and back ends, is intended for web-based deployment. Second, a rendering optimization method includes fire scenes and site codes. Lastly, a web-based serious game was developed to visualize the evacuation paths of multiple trainees, reveal evacuation regulations, and evaluate evacuation performance using evacuation paths and fire threats to provide valuable feedback. Trainees' evacuation abilities improve, proving the drill's usefulness. Drill findings showed that participants' evacuation abilities improved over time. Conclusions: (1) Virtual evacuation drills can be performed smoothly on the web using the designed serious game, and learners can receive rapid safety evaluation results. (2) The trainees' average evacuation time decreased during the practice. Trainees' safety scores rose. The results showed the game's usefulness. (3) Average evacuation efficiency and safety scores improved in the simulated practice with signs, suggesting that signals can help occupants evacuate. 4) The web-based serious game for virtual fire evacuation drills can improve trainees' safe evacuation abilities and aid in fire safety teaching. Web-based serious game virtual fire drills help improve evacuation abilities. Ruffino et al. [6] created a serious game to train people to evacuate a floor during a fire. Players learn the location of fire exits and firefighting equipment. Almeida et al. [16] used serious simulations to study building evacuations in emergency situations.

Most fire evacuation serious games are stand-alone programs, which limits their popularity. Stand-alone serious games are used for solitary training, making it harder to reveal numerous people evacuation. Web-based serious games must be developed to increase fire evacuation training participation, eliminating the requirement for downloading and installing. Online training can expose the evacuation characteristics of many students, guiding them to efficiently flee during a fire.

Niantic's Pokémon Go popularized location-based augmented reality games. The fact that players of these games perform actions in the external, physical world to accomplish in-game objectives motivates people to adopt them. They discuss disaster relief and the short- and long-term effects of AR games like Pokemon Go in catastrophe-prone locations. They report on initiatives to integrate knowledge about location-based AR games and

disaster management. They present a case study of a city with high natural disaster vulnerability – Wellington, New Zealand[42] – focused on describing the Pok emon Go gameplay mechanisms and emergency response plans. With this work, they set the stage for future empirical studies of location-based AR game players in urban areas and give policymakers with material to inform their handling of this emergent public interest dilemma. This study will ensure that augmented reality is not just integrated with the real world for gameplay and the provider’s revenue, but also for more effective crisis relief in a real disaster.

This paper examines how to use our youth’s entertaining yet worrisome habit of gaming for a higher purpose. Designing a game for disaster management can be used as a teaching tool to teach kids how to aid during a crisis. This game’s scenario is flash floods and cyclones. It helps countries prone to natural calamities and with less education. VR games have several uses. VR game for disaster relief [43]. VR games make players feel like reality, which is useful in various applications. This effort replicates a flood to teach disaster management. The game involves pre-flood measures. This game focuses on pre-disaster measures and rescue activities. This game can teach the uninformed what to do in a disaster. Kolb’s Learning Method focuses on the learner’s internal cognitive processes. After completing this game, its effectiveness is compared to disaster management books. This VR game allows players to make the appropriate decision before and during a calamity [43].

Serious games teach or train by authentically imitating a real-world setting and allowing learners to explore. In community policing, they can help LEAs and people adopt technologies[44], such as mobile and online apps, and enhance citizen awareness about community policing opportunities and foster citizen engagement. Community policing has three parts: Partnerships, transformation, problem solving. Police academies, local police, municipalities, and residents can all benefit from serious games. This article illustrates its application in the TRILLION project scenarios in five European cities (Lisbon, York, Ancona, Lecce and Eindhoven), which differ in population, geography, culture, and key security threats. In particular, TRILLION serious games are utilized to train and educate on community policing strategies and plans, with the aim of fostering dialogue and increasing subject alertness. They are also being used to improve the relationship between law enforcement agencies and citizens, particularly by promoting a more positive perception of law enforcement among those who view their authority as an infringement on their privacy. This approach is currently being trialed in various locations across Europe in close partnership with law enforcement agencies.

Our research aims to improve first-responder cybercrime training. We’ve tested serious games-based training materials using a Game Jam[45] and prototypes. The game poses a common police event, and the training mirrors the interaction with that person and the technological processes to preserve evidence for subsequent investigation. Many law enforcement agencies prioritize quick, cost-effective cybersecurity training for officers. Serious games can provide cheap, interesting, measurable training. They wanted to show that serious games can give updated training that is entertaining for the user, doesn’t pull cops off the streets for days at a time, and costs a fraction of traditional training methods. New social engineering types emerge quickly as employee attacks evolve. CyberSecurity Awareness Quiz gives an amusing quiz on recent variants to keep staff aware of new assaults or attack variants. Our part is to design questions and answers based on current events and crazy attacks. The CyberSecurity Knowledge Quiz is an online quiz to increase employee security awareness, especially against social engineering. The game uses real-world social engineering attacks as exam questions. New questions related to social engineering assaults will also be added. For this, a specific information-gathering process is designed.

This work presents a VR-based fire safety game. The game depicts a smoke-filled office, and the player must leave before the fire alarm sounds. The game was developed to analyze players’ activities as approximations of human behavior in a real fire. Fire education[46] has two goals: 1) Change people’s behavior to reduce fire mortality and morbidity; 2) Provide information to help people recognize dangers and respond to situations. Fire safety means preventing fires or restricting their spread. As fire safety issues are difficult to perform

in real life, VR apps are useful for teaching fire safety. Fire-related injuries are costly and slow to heal. This work introduces VirPa, a fire safety VR project. The study tested if a realistic VR game might be used for interactive fire safety learning and communication. It discusses empirically collected data on player behavior and user experience (UX) in an unexpected fire situation from the perspectives of fire safety and game design. Collecting player experiences helps evaluate VR's usefulness in the fire safety context, helping to future system design. Their results also suggest that playing serious games can raise fire safety awareness and improve players' abilities and understanding. The study protocol's hidden aim and the shift in game dynamics seem to have had a favorable effect. VirPa's realistic visuals and fire scenario gave players a fun learning experience. The game could emphasize fire safety to players. The game raised awareness about sudden fires and fire safety in general.

Military drills, CBRNE emergency training simulations, and strategy games to enhance abilities are examples of serious games (e.g., chess). Serious games range from basic card games to virtual reality and computer simulations. Serious games are increasingly utilized to study real-world situations, including floods. There has been limited synthesis of these approaches and their contributions to a more holistic and governance-based flood resilience viewpoint in flood risk management (FRM) [47]. Serious games offered an interactive and engaging method for addressing flood-related concerns. Flooding is a growing concern that is predicted to become more severe. Factors such as sea-level rise, increased occurrences of heavy rainfall, urban expansion and inadequate development are likely to raise the likelihood of devastating floods. To effectively manage flood risk, a more comprehensive approach is needed that considers all aspects of flood event, including preparation, response and recovery.

Natural disasters include earthquakes, typhoons, landslides, forest fires, tsunami, and floods. Serious games use amusement, creativity, and technology to teach, train, and enlighten. Serious games enable safe, cheap, engaging instruction. Serious games may be utilized for flood training. Malaysia floods annually. Civilians' understanding of flood preparation is minimal, and the current method doesn't meet APM's objectives[48]. This study analyzed flood awareness training preparation among Malaysians, the existing method for disseminating flood information, and the necessity for technology to carry out the training from an expert's standpoint. Flood training awareness is important because people must have duties in dangerous conditions. Physical and emotional preparation will reduce flood-related injuries and deaths. Serious games can be used to educate citizens about the necessity of flood training. Future studies will confirm the benefits of employing serious games for flood training.

A review of simulations serious games addressing disaster risk management (DRM) as tools for educating and engaging communities that are impacted, individuals with a vested interest and policy-makers is conducted in [49]. Recognizing the importance of a more dependable and evidence-based approach to evaluating the effectiveness of serious games in DRM, this research examines different types of activities and gathers both qualitative and quantitative data (such as player feedback, surveys, and scholarly articles on the games in question) to strengthen their evaluation. The potential and limitations of using games in the context of DRM are analyzed while also areas where game developers and researchers can create personalized game designs and valid evaluation methods are identified. DRM-related serious games/simulations provide a valuable social experience, where players work together to solve problems.. Serious games/simulations can reach broad audiences and realistically simulate disaster reality, which can help DRM raise disaster risk awareness, detect dangers, take preventative steps, trigger empathy, and gain perspective. Serious games may reach a wide audience and transmit trustworthy and consistent information on DRM, disaster awareness, hazards and vulnerabilities, and important skills throughout all stages of the DRM cycle.

Disasters have afflicted humans since birth, leaving them with painful memories. People have invented a number of ways to train disaster relief participants. Traditional training

Table 3. Disaster Management training approaches

Reference	Domain	Description
[27]	Flood awareness	A serious game to increase awareness on flood training preparation
[28]	Police training	TRILLION serious games are used to train and educate the community, stimulate discussion and create awareness
[40]	Disaster preventions	A serious game-based learning package for disaster prevention education
[41]	Fire Evacuation drills	A web-based serious game for virtual fire evacuation drills
[42]	Disaster relief	A location-based AR game on urban disaster management
[43]	Disaster relief	A serious game to get a better understanding of disaster management techniques
[44]	Cyber security survey	181 games related to cybersecurity
[45]	Police training	Game Jam will transform the delivery of first responder training in tackling cyber-crime
[46]	Fire Safety	An immersive and realistic VR game, suitable for interactive fire safety learning and communication
[47]	Flood risk management	A serious game to provide rationales such as learning, reflection, civic engagement, promoting behavior change as well as preparing players for future roles
[49]	Disaster risk management	A serious game for disaster awareness, portraying hazards and vulnerabilities, and modelling useful skills across all stages of DRM cycle
[50]	Disaster relief	To train participants in disaster relief and a large number of Serious Games for Disaster Relief (SGDRs)
[51]	Cyber security survey	A survey of 20 serious games for teaching cyber security at different levels

approaches, including disaster drills, may not fulfill today's needs. Many Serious Games for Catastrophe Relief (SGDRs) have been developed to train disaster planning and rescue [50]. This work proposed a thorough and professional disaster relief classification scheme. We discuss the purposes and technologies of serious games in each classification to help researchers understand and apply SGDRs.

Cybersecurity education is boosted by serious gaming. Serious games alter learners' knowledge, skills, and habits[51]. They compare 20 serious games for teaching cyber security on two levels. (2) the success of the games' organization design, instructional design and delivery, and game-based learning. The games' cyber security subjects are categorized using NIETP's cyber security knowledge units (KUs) for CAE/CDE-designated academic institutions. We rated the games' aesthetics, animation, audio, problem characteristics, and control ease. This study could help teachers choose good cyber security games. [52] found 181 cybersecurity-related games (either through content or aesthetics). Each game was played for an hour and its story, elements, and cybersecurity presentation were noted. Games and cybersecurity differed by platform and device (computer, mobile, or web). Most web games present cybersecurity as cyber safety and include quizzes or worksheets. Computer and mobile games present cybersecurity through aesthetics or deep content.

4. Discussion

Learning theories, game components, and learning results remain unclear. Learning outcomes depend on game aspects. Organizations must develop training goals and game components for engaging and meaningful gamified training. This research uses the TAM to show that at-risk employee behavior and information security awareness training are successful solutions. This study fills a literature gap by examining AI-enabled training. Some startups and hi-tech enterprises are using AI-enabled behavioral security training, which examines an employee's behavior for a few days or weeks before pushing them to

complete the training. This work will inspire future scholars to construct a SETA model integrating AI, Deep learning, and NLP for improved results. Serious gaming's long-term contribution to FRM is questionable. Future gameplay experiences must be explicit about their goals and objectives and follow up with participants to identify long-term retention and impacts. Although serious games are popular, additional study is needed to assure their usefulness and help people comprehend vital subjects. Natural catastrophe training uses serious games.

To make the game more exciting, users should control characters, click stuff, etc. We intend to find a suitable learning material structure to improve enjoyment in the designed game by utilizing game theories on assessing game enjoyment, such as game refinement theory. Most VR serious games lack realistic NPCs; future developments should address this. Little is known about VR serious games' ability to improve rescuers' skills. This article: Thus, a low-cost, extremely immersive virtual reality rescue training serious game with 3D character models is created. With the variety of digital cybersecurity games and platform variances, game makers should focus on cyber safety rather than gamification. The present scope of cybersecurity games allows for deeper content interaction with cybersecurity themes in an environment conducive to widening cybersecurity participation.

Developers should research platform usability and improve it. They should also examine what causes software engineers to grasp the ramifications of exploiting weak code during CyberSecurity Challenges. In quizzes, an overarching storyline could be included to make the experience more immersive and add more actions and components from game refinement theory or other frameworks to boost satisfaction. Chatbot needs many upgrades. The chatbot should have more features and routines to help people. The chatbot can be integrated into personal assistant apps to warn about cybersecurity threats. The chatbot's knowledge base should also include less-related cybersecurity terms. Future works With current approaches, civilians aren't aware of the need of flood training. Flood victims must learn and practice preparations to be ready for a flood and know what to do.

The following best practices should be implemented:

- Establish a culture and participation throughout the organization in which cybersecurity decision-making and implementation are integrated into the daily tasks of all end-users.
- Clearly communicate to all users, regardless of their organizational position, the importance and purpose of cybersecurity education before initiating training.
- Assess program achievement by performing a comparison analysis to assess if institutional employee-driven cybersecurity events are decreasing over time.
- Conduct regular, ongoing evaluations and training to provide end-users with regular cybersecurity education as well as the ability to learn and build new abilities over time.
- Make a clear connection between assessments and training.
- Keep end-users informed of the cyber-security subject by regularly going over it and implementing ongoing awareness initiatives.; without reinforcement, the institution will have to reconstruct rather than build.
- Track and report progress on a regular basis.
- Maintain end-user motivation and engagement by implementing gamification approaches that utilize rewards and positive feedback to increase users interest and involvement and improve program effectiveness.

5. Conclusions

Game elements in ICTs are a proven approach to engage users. Many consumers enjoy gamified work in several fields. Privacy should be considered when building them because they record personal information and actions. Several approaches examine security and privacy requirements, but few integrate them with gamification. In this work, privacy and gamification are examined, and gamification may compromise privacy. Users must understand privacy and security to defend themselves. These two factors need education

initiatives. Gamification makes user training more engaging. Attempts were made. Privacy awareness can be achieved by building gamified systems that teach users how to safeguard their privacy from harmful game features and the penalties of privacy infringement. So, when using gamified systems, users will know which game features to pay attention to safeguard their privacy and how to avoid their adverse effects. Future work will promote security- and privacy-friendly software patterns. Gamification will be used to study user privacy and security. The relationship between gamification, security, and privacy must be evaluated from both the developer and user perspectives.

Cyber security is a fast emerging topic with growing societal influence and vast, sophisticated data sets on networked computing devices. Most crimes investigated by Police Scotland use computer technology, thus training in this area is crucial. Classroom training is costly and time-consuming. Serious games can instruct police officers about cybersecurity. Our project prototype fuses cyber security with computer gaming technologies to create an interactive training tool. This tool has shown that effective training can be offered at a lower cost, to more staff, and without costly classroom sessions. Increasing law enforcement cyber-security expertise will improve cyber-crime response and digital evidence preservation.

These studies highlighted the necessity for educational content that may successfully fill the gap of multiple playing, and we want to develop and measure such content in the future. In recent years, industrial control system and cybersystem security incidents have increased. Several incidents were caused by inadequate software development practices, which may be due to a lack of secure coding knowledge among developers. Lack of cybersecurity alertness is a problem for critical infrastructures since exploiting vulnerable code can lead to loss of life. Serious games can enhance software developers' understanding of secure coding and principles.

In conclusion, it is essential that a comprehensive cybersecurity awareness program is implemented within organizations, with management setting a leading example for all users to follow. This program should not only inform and educate users on the company's security policies, but also keep them updated on any changes or updates for compliance purposes. Furthermore, it is crucial that employees are made aware of their responsibilities and that there will be consequences for non-compliance. By ensuring that staff is well-informed about the organization's security risks and expectations, the overall level of security will be significantly enhanced.

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Abbreviations

The following abbreviations are used in this manuscript:

AR Augmented Reality
MDPI Multidisciplinary Digital Publishing Institute
SG Serious Games
VR Virtual Reality

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